

Investigating Discrete Guidance Control in CFG-Ctrl

Arthur Gislon Leonida | Ongoing independent research

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Original paper and proposed solution. Wang et al.’s *CFG-Ctrl: Control-Based Classifier-Free Diffusion Guidance* (2026) interprets classifier-free guidance (CFG) as proportional feedback on the conditional–unconditional velocity discrepancy, $e = v_c - v_u$. The authors propose sliding-mode guidance (SMC-CFG) to address overshooting at high guidance scales, with a switching correction and a continuous-time Lyapunov analysis under stated assumptions. They report improved alignment and robustness on SD3.5, FLUX and Qwen-Image; those findings are not independently established here.

Problem identified in the reimplement. The implemented discrete controller stores the corrected discrepancy:

$$s_n = e_n + (\lambda - 1)(e_{n-1} + \delta_{n-1}), \quad \delta_n = -k \operatorname{sign}(s_n), \quad \hat{v}_n = v_{u,n} + w(e_n + \delta_n).$$

Late in sampling, small current and previous discrepancies leave $s_n \approx (\lambda - 1)\delta_{n-1}$: the controller can react mainly to its own previous correction. After a nonzero correction, $\lambda = 6$ and $k = 0.1$ permit the componentwise cycle

$$\delta_{n-1} = +0.1 \implies s_n \approx +0.5 \implies \delta_n = -0.1 \implies s_{n+1} \approx -0.5 \implies \delta_{n+1} = +0.1.$$

Thus s can oscillate across zero rather than settle at $+0.5$; its unsigned RMS stays near 0.5 even if its signed average is near zero. Historical runs showed surface RMS 0.509–0.515 with error RMS near 0.01 and almost complete sign switching. The correction remains near ± 0.1 , and the sampler receives $w\delta_n$. The paper acknowledges large-gain chattering; this project identifies a specific corrected-memory mechanism, whose image-quality impact remains open.

Alternatives implemented so far. Initial variants use measured-error memory and a smooth saturation switch, correcting only CFG’s extrapolation beyond the conditional prediction. Their surface is $s_n = e_n + (\lambda - 1)e_{n-1}$, so it approaches zero if both measured errors do, without recycling the previous correction.

The main proposed alternative replaces switching feedback with memoryless shrinkage: retain the conditional prediction and attenuate the extra CFG extrapolation according to the *current* discrepancy. For $w \geq 1$,

$$T_\tau(e) = \operatorname{sign}(e) \max(|e| - \tau, 0), \quad \hat{v} = v_c + (w - 1)T_\tau(e).$$

Each component loses at most its own magnitude: values below τ contribute no extra extrapolation, while larger values retain their sign. **proximal** uses $\tau = k$; **proximal_relative** uses $\tau = k \operatorname{RMS}(e)$ per sample. For a scalar $e = 0.01$ and relative $k = 0.3$, $\tau = 0.003$ and $T_\tau(e) = 0.007$: the adjustment scales with the remaining signal. Relative k controls shrinkage strength, not a fixed push. The change from ordinary CFG satisfies $|\Delta v_i| \leq (w - 1)|e_i|$, so it vanishes as $e \rightarrow 0$.

These proximal variants have *no sliding surface*: the logged $s = e$ is only a diagnostic reference. Its RMS follows the current error rather than plateauing at $(\lambda - 1)k$. A finite sampling run need not end at zero: **rel03** logged final error/reference RMS 0.02749, measured before the last scheduler update. Memory-induced oscillation is removed, but changing model predictions can still change signs. No extra model calls are required; better images are not guaranteed because shrinkage can remove useful details.

Preliminary image results. A development run used SD3.5-Large, 64 COCO prompts, one image per prompt and $w = 7.5$, with paired prompts and seeds across methods. For relative gain $k = 0.3$ (**rel03**), the saved report and supplied comparison log give:

Metric	Candidate mean	Difference from SMC-CFG	Reported 95% CI
CLIP	0.27303	−0.001865	[−0.006385, 0.002426]
PickScore	0.23122	−0.000957	[−0.001842, −0.000042]

Both metrics are normalized image–text cosine scores; higher is better. CLIP is inconclusive, while the nominal PickScore interval favours SMC-CFG. Comparisons with ordinary CFG and relative $k = 0.1$ span zero for both metrics. This exploratory run establishes no benefit from increasing relative k to 0.3.

Ongoing work. Next are paired gain/scale sweeps with equally tuned baselines, held-out prompts and more seeds, then the paper’s COCO, T2I-CompBench and GenAI-Bench metric families. Full evaluation remains outstanding; unspecified paper settings limit exact reproduction. The question is whether improved controller behaviour yields better image quality at comparable prompt alignment.

Reference: H. Wang, Y. Liu, J. Chi, F. Liu, R. Xue and Y. Duan (2026), *CFG-Ctrl: Control-Based Classifier-Free Diffusion Guidance*. arXiv:2603.03281v2.